

DATASET ASSESSMENT DOCUMENT (DAD)

1. Dataset Overview

Item	Details
Dataset Title	Manufacturing Operations & Performance Dataset
Domain / Theme	Manufacturing Operations, Production Analytics, Quality & Reliability
Source	Chat GPT
File Type	CSV (Raw), Excel (Cleaned & Normalized)
No. of Rows	7,600 (Raw Dataset)
No. of Columns	31 (Pre-Normalization)
Data Structure	Star Schema (Fact + Dimension Tables)

2. Data Structure Summary

Fact Table

- Fact_Production

Dimension Tables

- Dim_Line
- Dim_Plant
- Dim_Product
- Dim_Shift
- Dim_Machine
- Dim_Supplier
- Dim_ScrapReason
- Dim_DowntimeReason
- Dim_MaterialLot
- Dim_QAResult
- Dim_Supervisor

3. Column Classification Summary

Column Name	Data Type	Dimension / Measure	Description
Batch Date	Date	Dimension	Production date
Plant City	String	Dimension	Manufacturing plant location
SKU	String	Dimension	Product SKU identifier
Product Category	String	Dimension	Product classification
Shift	String	Dimension	Production shift
Machine ID	String	Dimension	Machine identifier
Supplier Name	String	Dimension	Raw material supplier
Production Units	Integer	Measure	Total produced units
Good Units	Integer	Measure	Accepted output units
Scrap Units	Integer	Measure	Rejected units
Downtime Minutes	Integer	Measure	Total downtime duration
Energy Consumption	Decimal	Measure	Energy used (kWh)
Water Consumption	Decimal	Measure	Water used (Liters)
QA Score	Decimal	Measure	Quality assessment score
Plan Output	Integer	Measure	Planned production output

4. Data Quality Assessment

Aspect Checked	Observations	Action Taken
Missing Data	Present in certain operational fields	Imputed or validated logically
Duplicates	No major duplicate batch records	Verified uniqueness
Outliers	Present in Scrap Rate & Downtime	Evaluated during analysis
Incorrect Data Types	Text stored as numeric in some fields	Converted to proper formats
Consistency Issues	Standardization required for categorical fields	Cleaned & standardized

5. Key Metrics & KPIs Identified (Initial)

KPI	Why It Matters
Plan Attainment %	Measures production efficiency vs target
Yield %	Indicates manufacturing effectiveness
Scrap Rate %	Measures quality loss
Downtime %	Measures operational inefficiency
Energy per Good Unit	Sustainability efficiency metric
Water per Good Unit	Resource optimization metric
QA Score	Product quality benchmark

6. Assumptions / Risks

- Batch granularity assumed at production-run level.
- QA score assumed standardized across plants.
- Supplier quality impact inferred from scrap and QA trends.